



基于火灾危险气体释放与风场驱动二维对流-扩散模型的校园室外疏散方案设计方法： 以一零一中学为例

**Design Method of Campus
Outdoor Fire Evacuation Plan
Based on Hazardous Gas
Release and Wind-Driven 2D
Advection-Diffusion Model:
*A Case Study of Beijing 101 Middle School***

Process Report: Stage 1
(May. Week 4)

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Chapter 1. Timeline Review 时间计划安排回顾

时间线 Timeline



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最终提交时间 Final Deadline

本项目计划在**2026年6月第三周**完成最终递交。

The project is planned to be completed in the third week of June 2026.

第一阶段：资料整理与模型确定 | May Week 4

完成研究背景、化学原理、危险气体类型、二维对流-扩散模型和校园地图抽象方法的整理，明确研究假设与简化方案。

Collect background information, combustion chemistry, hazardous gas types, the 2D advection-diffusion model, and the simplified campus map representation.

第二阶段：模型计算与方案分析 | June Week 1

建立火源、风向、扩散范围、路径风险和集合点评价方式，比较不同情景下的疏散路线安全性。

Build fire source, wind direction, diffusion, path risk, and assembly point evaluation models, then compare evacuation safety under different scenarios.

第三阶段：论文、海报与视频制作 | June Week 2

完成论文初稿、海报主要版面和视频脚本，将化学模型、数学公式、地图图示和疏散方案整合为可展示成果。

Complete the paper draft, poster layout, and video script, integrating chemistry, mathematical formulas, campus maps, and evacuation plans.

第四阶段：修改、排版与最终提交 | June Week 3

检查逻辑完整性、双语表达、图表引用和格式规范，完成最终版本并提交论文、海报和视频。

Revise logic, bilingual wording, citations, figures, and formatting before submitting the final paper, poster, and video.



Chapter 2. Completed Content 完成内容

Contents

1	Introduction	2
2	Notation and Parameters	2
3	Model Assumptions	4
4	Overall Model Framework	5
5	Data Sources and Preprocessing	6
6	Model 1: Hazardous Gas Generation Model	6
7	Model 2: Wind-Driven 2D Advection-Diffusion Model	7
8	Model 3: Evacuation Cost and Route Optimization Model	7
9	Case Study and Simulation Results	7
10	Sensitivity Analysis	7
11	Conclusions	7
12	Discussions	7
13	Acknowledgments	7
	Appendices	7

Ch1. Introduction



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Ch2. Notation and Parameters Table



Notations and parameters in this research are shown below.

Table 1: Notation and Parameters

Symbol	Definition	Unit / Type
Notation for Field and Discrete Field		
(x, y)	Continuous spatial location	m
(i, j)	Grid-cell index in the discrete field	—
t	Continuous time	s
n	Discrete time-step index	—
$\Delta x, \Delta y$	Grid spacing in the x and y directions	m
Δt	Time-step size	s
$\vec{w}(x, y)$	Continuous wind velocity field	m/s
$\vec{w}_{i,j}$	Wind vector at grid cell (i, j)	m/s
$u(x, y), v(x, y)$	Continuous wind components	m/s
$u_{i,j}, v_{i,j}$	Discrete wind components at cell (i, j)	m/s
$C_g(x, y, t)$	Concentration field of gas g	kg/m ³
$C_{g,i,j}^n$	Discrete concentration at cell (i, j) and time step n of gas g	kg/m ³
Geographical Information and Road Network Graph		
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Campus road network graph	graph
$\mathcal{V} = \{v_1, v_2, \dots, v_N\}$	Set of nodes	nodes set
$\mathcal{E} = \{e_1, e_2, \dots, e_M\}$	Set of edges	edges set
v_k	The k -th node	node
id_k	ID of node v_k	integer
(x_k, y_k)	Position of node v_k	m
$\text{type}(v_k)$	Type of node v_k	categorical
P_k	Population assigned to node v_k	persons
K_k	Capacity of node v_k , if applicable	persons
$e_m = (v_a, v_b)$	The m -th edge connecting v_a and v_b	edge
$d_m = \text{dist}(e_m)$	Distance of edge e_m	m
K_m	Capacity of edge e_m	persons
Model 1: Hazardous Gas Generation		
i	Index of hazardous gas species	—
M_b	Type of building material	categorical
q_i	Average generation rate of gas i	kg/s
t	Time after ignition	s
$m_i(t)$	Mass of gas i generated by time t	kg
A_s	Effective source area in the 2D model	m ²
V_s	Effective source volume, if using mass concentration	m ³

(Continued table)

Symbol	Definition	Unit / Type
$C_{s,i}(t)$	Source-region concentration of gas i	kg/m ³
$S_i(t)$	Source term of gas i	kg/(m ² s)
Model 2: Wind-Driven 2D Advection-Diffusion		
$C_g(x, y, t)$	Concentration of gas g	kg/m ³
$D_{\text{eff},g}$	Effective diffusion coefficient of gas g	m ² /s
$\vec{w}(x, y) = \langle u, v \rangle$	Wind velocity vector field	m/s
u	Vector component in x direction	m/s
v	Vector component in y direction	m/s
$S_g(t)$	Source term from Model 1	concentration/s
λ_g	Attenuation coefficient of gas g	s ⁻¹
$\frac{\partial C_g}{\partial t}$	Rate of concentration change over t	concentration/s
$\frac{\partial C_g}{\partial x}$	Rate of concentration change over x	concentration/m
$\frac{\partial C_g}{\partial y}$	Rate of concentration change over y	concentration/m
$\nabla^2 C_g$	Diffusion-related spatial curvature of concentration	concentration/m ²
Model 3: Evacuation Cost and Route Optimization		
k	Index of cluster of people	—
h	Index of scheme for cluster k	—
$\mathcal{P}_k$	Set of feasible evacuation paths for cluster k	path set
$p_{h,k} \in \mathcal{P}_k$	A candidate full path for cluster k	path
$L(p_k)$	Total length of path p_k	m
$E(p_k)$	Exposure along path p_k	concentration · s
$G(p_k)$	Congestion level along the most crowded path p_k	dimensionless
s	The speed of crowded	m/s
$T(p_k)$	Total time consume of path p_k	s
L_0, E_0, G_0	Normalization parameters	corresponding units
α, β, γ	Weights in the cost function	dimensionless
$\text{Cost}(p_k)$	Total cost of candidate path p_k	dimensionless
p_k^*	Optimal path for cluster k	path

3 Model Assumptions

To control the scale of the study, appropriate assumptions are made for the model:

1. This study only discusses the outdoor evacuation process on campus after personnel have left the building. The building on fire is simplified as a source of dangerous gas release, with the release intensity estimated based on the type of burning material, burning rate, and gas generation coefficient;
2. During a short-term evacuation process, the impact of outdoor gases is primarily concentrated near the burning building, around exits, on downwind roads, and at assembly points located closer to the fire source;
3. The actual simulation employs the finite difference method to represent and calculate physical fields using grids;
4. Campus roads, building exits, and assembly points will be abstracted into a graph structure, with road segment capacity and assembly point capacity treated as known or estimable parameters in the model;
5. Evacuation behavior is handled based on population units, without further breakdown into individual complex behaviors for the time being.

Ch4. Model Framework

4 Overall Model Framework

The overall framework of this study is a connected models chain: begin at fire chemistry, going to hazardous gas diffusion, and finally evacuation scheme optimization.

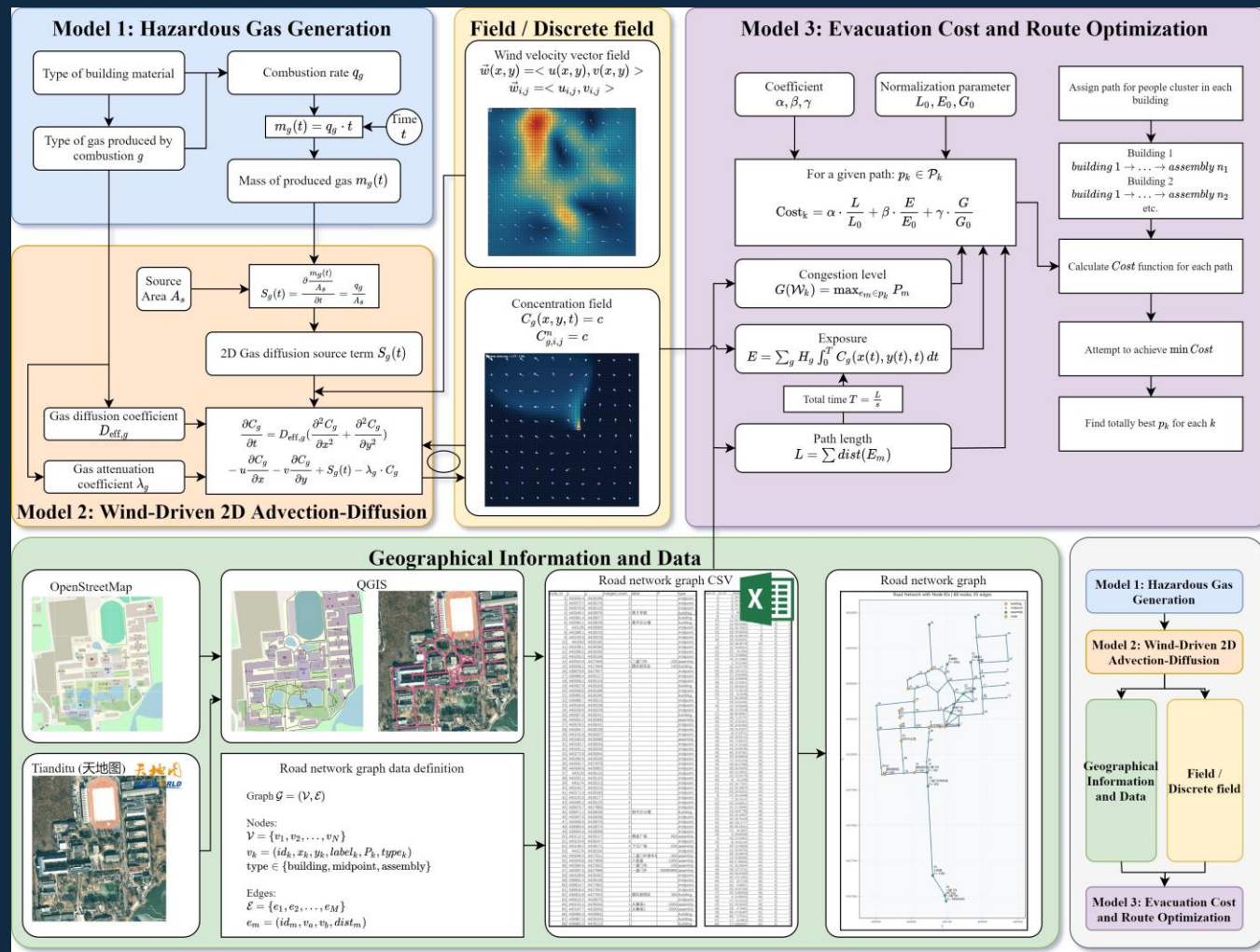
First, model 1 simplifies the combustion process of building materials into a hazardous gas generation function, which estimates the mass $m_g(t)$ of gas species g generated by time t .

Second, model 2 convert mass function into the source term $S_g(t)$ that used in the diffusion model. Then, $S_g(t)$ is inserted into a two-dimensional advection-diffusion equation with wind influence based on Fick's law. This model describes the outdoor spread of hazardous gas on campus using the continuous concentration field $C_g(x, y, t)$, which is discretized into grid-based concentrations $C_{g,i,j}^n$ through a finite-difference method, and the size of each grid are define as Δx and Δy . The wind field in the diffusion equation is represented as a two-dimensional vector field $\vec{w}(x, y) = \langle u(x, y), v(x, y) \rangle$, and in the discrete grid it is denoted as $\vec{w}_{i,j} = \langle u_{i,j}, v_{i,j} \rangle$. The main output of Model 2 is the time-dependent concentration distribution.

Thirdly, model 3 combines the concentration field produced by Model 2 with the real campus road network graph. Buildings, road turns, and assembly points are abstracted as nodes v_k , and main walkable roads are abstracted as edges, forming the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. For some nodes which are people cluster k , the model compares candidate paths p_k within the feasible path set $\mathcal{P}_k$. The cost of a path depends on its total distance $L(p_k)$, total exposure value $E(p_k)$, bottleneck congestion level $G(p_k)$. Therefore, the evacuation problem is transformed into a comprehensive cost minimization problem, and able to solve by algorithms such as Time-space dependent A* or dynamic Dijkstra.

This variable chain shows how the chemical part, diffusion part, and evacuation optimization part are connected. The final output p_k^* represents the recommended evacuation path for people cluster k , selected by minimizing the total cost among feasible paths. In this way, the project links AP Chemistry, AP Calculus BC, and computer science into one integrated modeling framework.

$$t \rightarrow m_g(t) \rightarrow S_g(t) \rightarrow C_g(x, y, t) \rightarrow C_{g,i,j}^n \rightarrow E(p_k) + L(p_k) + G(p_k) \rightarrow \text{Cost}(p_k) \rightarrow p_k^* \quad (1)$$





Thanks!

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